Textual Signals from SEC 8-K Filings: Incremental Predictive Value Beyond Volatility in the AI Sector

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Abstract

Since the public launch of ChatGPT in November 2022, firms across the AI value chain have increasingly described AI demand, GPU capacity, and capital expenditure in their earnings disclosures. To investigate whether this language, extracted from SEC 8-K Item 2.02 filings for a 19-ticker AI-sector universe, adds predictive value beyond a volatility-only baseline for cross-sectional stock ranking, we combined NLP features with yfinance price data (2019–2025) in a LightGBM model. Filing-level signals extracted from SEC EDGAR, such as a custom AI-sentence ratio, GPU mentions, capex, and competitive language, were combined with price and used to predict which AI-sector stocks are expected to outperform their peers over the next five trading days. As a robustness check, we also added headline sentiment from FNSPID, a financial news dataset, which did not strongly impact the results. The model was evaluated using walk-forward cross-validation across five expanding folds with a 21-day purge. It achieves a mean Information Coefficient (IC) of 0.0185 and an IC Information Ratio (ICIR) of 0.6671, but relative to a volatility-only model, the text features add no significant predictive value ($\Delta IC = 0.008$, Newey-West $p = 0.57$). Additionally, a leakage analysis demonstrates shifting the label window to overlap the feature date by even one trading day produces highly significant ICs (+0.008 per day of overlap, p=0.012), suggesting that AI-related disclosure language is full of noise that makes ranking stock performance difficult, and that apparent signals should be scrutinized for lookahead bias.

Code: GitHub repository
Demo: Streamlit dashboard

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1 Introduction

Following the release of ChatGPT in 2022, generative artificial intelligence (AI) has rapidly permeated technology in nearly all industry sectors. In 2025, over 60 % of the total capex spent by the four largest hyperscalers (Amazon, Microsoft, Google, and Meta, at $156B out of $260B collectively) went towards AI, including GPUs and data centers. (10) This raises the question for investors: *As firms increasingly describe AI demand, GPU capacity, and capital spending in their earnings disclosures, does that language contain additional information about future relative returns that prices and volatility do not already contribute?*

We use SEC 8-K Filings, which include updated reports that provide crucial financial information on material events of a business, including revenue guidance and unaudited quarterly financial statements (11). Specifically, we focus on Item 2.02 since these documents are furnished rather than filed (Exhibit 99.1); they face less legal scrutiny than audited reports and would thus more accurately reflect how management aims to depict their results. The primary advantages of using these filings for systematic analysis are that they are mandated, timestamped, and consistently structured across firms.

The second concern is methodological. Financial machine learning results are especially prone to lookahead bias, where information that should be unavailable at the predicted time is used in features, potentially incorrectly forming large, statistically significant results. In our own pipeline, an off-by-one error in constructing the five-day forward-return target initially yielded an IC of 0.061, which disappeared once the target was corrected. We therefore treat leakage as a research question in its own right and measure how much it inflates results.

This motivates two research questions:

RQ1. Do NLP signals extracted from SEC EDGAR 8-K filings, specifically AI demand language, mentions of GPUs, and capital expenditure signals, add measurable predictive value beyond a volatility-only model for cross-sectional stock ranking in the AI sector?

RQ2. How sensitive are cross-sectional IC estimates to lookahead leakage in target construction, and does inflation scale with the degree of overlap between the label window and the feature date?

This paper makes three contributions. First, we construct dictionaries with AI-related terminology covering demand, GPU, capital expenditure, and competitive language that semiconductor companies commonly use. Second, we evaluate these signals in a leakage-controlled walk-forward evaluation against a volatility baseline and find no significant incremental value. Third, we quantify how lookahead leakage inflates the information coefficient, showing that inflation grows with each day of overlap between the label window and the feature date.

1.1 Related Works

Loughran and McDonald note that the Harvard Psychosociological Dictionary, commonly misclassifies about 75% of words as having negative sentiment when drawn from a financial context. Although they developed an alternative lexicon that better reflects negative financial sentiment, their dictionary precedes the Generative AI revolution and does not contain AI-related vocabulary (7). Instead, we construct dictionaries that specifically focus on AI demand, GPU, capital expenditure, and competitive language.

Gu, Kelly, and Xiu find that tree ensembles and neural networks outperform linear models in predicting cross-sectional returns because they capture nonlinear interactions among predictors, and that volatility is among the most informative predictors (6). We therefore use LightGBM, a gradient-boosted tree ensemble. Our panel of 29,487 stock-days across 19 tickers is far smaller than their universe of tens of thousands of stocks, and at this scale tree-based models are generally more reliable than neural networks on tabular data.

2 Methodology

2.1 Data and Features

To build a machine learning pipeline that predicts which AI sector stocks would outperform their peers over a five trading day horizon, financial data from yfinance (Yahoo Finance) was collected from 2019-2025 in daily intervals across 19 tickers. These tickers span four buckets of the AI value chain: semiconductors, hyperscalers, AI pure plays, and infrastructure (Table 1). The segments describe the composition of the 19-ticker universe, with the prediction target ranking each stock against all others.

Filing text comes from SEC EDGAR 8-K text filings under Item 2.02, which contains quarterly earnings releases. Using Regex, we remove HTML tags and entities, as well as excess whitespace, monetary values, and special characters. We also omit dates and articles shorter than 200 characters. As a robustness check, we score all FNSPID news headlines (5) for each ticker from 2019-2023. We applied the FinBERT model (1), a BERT-based language model that is fine-tuned on financial text, producing daily mean sentiment, article count, and share of positive headlines. These features are excluded from the main model.

Features were primarily created based on two groups:

Table 1: Ticker classification by AI-sector exposure category.

Bucket	Tickers	Definition
Semiconductors	NVDA, AMD, INTC, ASML, TSM, AVGO, ARM, MRVL	Designs, manufactures, and sells microchips for electronic devices
Hyperscalers	MSFT, GOOGL, AMZN, META	Invests capital in building and expanding data centers; provides scalable cloud computing infrastructure
AI Pure Plays	PLTR, AI, SOUN	Generates the majority of its revenue from AI tech, products, or services
Infrastructure	DELL, ANET, HPE, SMCI	Specializes in hardware, server, and high-speed networking

- **Price and Volatility:** Features such as `vol_5d` and `vol_20d` kept track of the volatility (rolling standard deviation) over n days. `vol_spike` showed the volume relative to the average over a 20-day rolling window. Any columns involving returns were excluded to avoid lookahead bias. Fundamentally, `fwd_5d` was excluded because it was used to create the Target variable.
- **Filing Signals:** Dictionaries with domain-specific terminology related to NVIDIA, AMD, Intel, and more, were defined involving words indicating GPU demand, whether capex has gone up or down, and risk words related to competitive risk. These were used to create a metric that identified the ratio of sentences related to artificial intelligence relative to the total number of sentences.

For each filing f of stock i, the score for dictionary c and the AI-sentence ratio are

$$s_{i,c,f} = 1000 \times \frac{\sum_{k \in \mathcal{K}_c} \text{count}(k, D_{i,f})}{W_{i,f}},$$

$$a_{i,f} = \frac{\left|\{\text{sentences in } D_{i,f} \text{ with an AI phrase}\}\right|}{\left|\{\text{sentences in } D_{i,f}\}\right|},$$

where $\mathcal{K}_c$ is the set of keywords in dictionary c (full lists in the code repository), $\text{count}(k, D_{i,f})$ is the number of case-insensitive occurrences of keyword k in filing text $D_{i,f}$, and $W_{i,f}$ is the filing's word count. Net capital-expenditure language is $s_{i,\text{capex_up},f} - s_{i,\text{capex_down},f}$. Because earnings releases are typically filed after the market close, filing scores are mapped to trading days with a one-day lag,

$$x_{i,c,t} = s_{i,c,f^*(t)},$$

where $f^*(t)$ is the most recent filing with filing date $\leq t-1$, so each score is carried forward until the firm's next filing and is set to 0 before its first filing.

The target is cross-sectional. The forward five-day return $R_{i,t} = \sum_{h=1}^{5} r_{i,t+h}$ sums the daily simple returns of stock i over the next five trading days, and the label compares it with the median across the universe $\mathcal{N}$ on the same day:

$$y_{i,t} = \mathbf{1}\left[R_{i,t} > \underset{j \in \mathcal{N}}{\text{median}} R_{j,t}\right].$$

Each stock is labelled by comparing its forward five-day return with the returns of all stocks on the same date. We chose the cross-sectional ranking over the absolute direction prediction to eliminate bull market bias. As of July 2026, the United States was officially in a matured bull market driven by hyperscalers and technology companies investing hundreds of billions of dollars into semiconductors and data centers (9), so an absolute direction model could largely succeed by predicting that every stock rises. Comparing a stock's performance with the median five-day cumulative return of that date also yields an approximately 50/50 class balance. Observations without a complete five-day forward window were excluded.

2.2 Model and Validation

This project uses the **LightGBM Classifier** with the following hyperparameters: `n_estimators=600`, `learning_rate=0.05`, `num_leaves=31`, and `class_weight='balanced'`. The model used early stopping after 50 rounds to prevent overfitting. For walk-forward validation, the Scikit-learn package `TimeSeriesSplit` was used to create five expanding folds (Table 2). Since the label at day t spans returns from t+1 to t+5, a gap longer than 5 days ensures no training label overlaps the test period. The parameter `gap_days` was set to a default of 21 days (the number of trading days in a standard month) to prevent lookahead bias while forward-filling. Within each train set, the last 15% of dates were held out due to early stopping.

2.3 Evaluation Metrics

The Information Coefficient (IC) is the cross-sectional Pearson correlation between predicted outperformance probabilities $f_{i,t}$ and realized labels $y_{i,t}$:

$$IC_t = \frac{\sum_{i=1}^{N}(f_{i,t} - \bar{f}_t)(y_{i,t} - \bar{y}_t)}{\sqrt{\sum_{i=1}^{N}(f_{i,t} - \bar{f}_t)^2 \sum_{i=1}^{N}(y_{i,t} - \bar{y}_t)^2}},$$

where $N = 19$. Because the label is binary, this is a point-biserial correlation. Unlike directional accuracy, it uses the full predicted probability, rewarding confident correct rankings over marginal ones. The IC Information

Table 2: Walk-forward results by fold ($k = 0$, correctly constructed target).

			IC		
Fold	**Train end**	**Test period**	**Model**	**Base.**	**Δ**
1	Feb 2020	Mar 2020–Apr 2021	−0.021	0.002	−0.023
2	Mar 2021	Apr 2021–Jun 2022	0.053	0.008	0.045
3	May 2022	Jun 2022–Aug 2023	0.023	0.002	0.021
4	Jul 2023	Aug 2023–Oct 2024	0.007	0.005	0.002
5	Sep 2024	Oct 2024–Dec 2025	0.031	−0.008	0.040
Mean			0.019	0.002	0.017

Note: Training begins 2019-01-02. ΔIC = Model IC − Baseline IC. Baseline IC is the correlation of raw `vol_5d` with the label. Test periods are separated from training data by a 21-trading-day gap.

Table 3: LightGBM feature importance (number of splits) in the final-fold model.

Feature	**Splits**	**Share (%)**
`vol_20d`	12	40.0
`nvidia_mentions`	4	13.3
`vol_spike`	3	10.0
`capex_up_score`	3	10.0
`competitor_mentions`	3	10.0
`ai_sentence_ratio`	3	10.0
`amd_mentions`	2	6.7
Total	30	100.0

Note: `vol_5d`, `gpu_mentions`, `capex_net`, `capex_down_score`, and `intel_mentions` were not used in any split.

Ratio, ICIR $= \mu_{IC}/\sigma_{IC}$, measures the consistency of IC across folds. With only five folds, we report it descriptively. For inference, we compute IC on each of the 1,455 test days and test whether its mean is zero using Newey–West standard errors (20 lags), which correct for the autocorrelation created by overlapping five-day labels.

We compare the full model against two benchmarks. The *raw* `vol_5d` *baseline* is the correlation between `vol_5d` (the rolling volatility over a 5-day window) and the label in each test fold, with no model trained. The *volatility-only model* is a LightGBM model trained on `vol_5d`, `vol_20d`, and `vol_spike` (volume spike in shares traded on a given day) with the same folds and hyperparameters. Volatility was selected for the baseline model because Gu, Kelly & Xiu (2020)'s prior work identifies it as a dominant predictor signals for estimating cross-sectional return behavior (6).

2.4 Leakage Analysis

To measure the sensitivity of our results to lookahead leakage, we repeat the analysis with deliberately misaligned targets. We constructed windows from time $t+1-k$ to $t+5-k$, where t is the current date and k is the number of trading days the label window overlaps the feature date. $k = 0$ is equivalent to having no error in leakage. We held the same features, hyperparameters, folds, purge, and seed constant, only changing the label window. For each k, we test whether the mean daily IC exceeds zero and whether IC increases linearly with k.

3 Results

3.1 Walk-Forward Performance

The model achieves a mean fold IC of 0.0185 and a fold ICIR of 0.67 (Table 2). Across all 1,455 test days, the mean daily IC is 0.015 (daily ICIR 0.06), which is not significantly different from zero (Newey–West $t = 1.34$, $p = 0.18$).

The LightGBM model outperforms the baseline in 4 of 5 folds. However, the differences are small and not statistically significant. The baseline model was a single untrained feature (the raw `vol_5d` feature). Using a binomial test distribution, the probability of at least 4 wins in 5 folds would occur by chance is about 0.19. The text features contribute a mean daily $\Delta IC = 0.008$ relative to the volatility-only baseline, which has no statistical difference from 0 (Newey–West p=0.57). In other words, in this study, the text does not add any significantly detectable signals. In a robustness check that adds FNSPID news sentiment to the main model, the text features still do not have significant predictive value and change this conclusion.

Fold-level IC ranges from −0.021 (Fold 1, March 2020 – April 2021) to 0.053 (Fold 2, April 2021 – June 2022). Given roughly 290 test days per fold and a daily IC standard deviation of about 0.25, each fold's IC carries a standard error of approximately 0.015, so variation of this size across folds is consistent with sampling noise.

3.2 Feature Importance

Consistent with Gu et al.'s finding that volatility is a strong cross-sectional predictor, `vol_20d` ranked the highest, accounting for about 40% of all tree splits. Filing-based NLP features make up the rest of the most impactful features, predominantly led by `nvidia_mentions`. However, the model only consisted of 30 tree splits in all, indicating these rankings reflect a small number of decisions and should be interpreted cautiously.

In Figure 1, `vol_20d` has the highest spread but no clear direction. High (red) and low (blue) values appear on either side of 0.00 Shap value, indicating both positive and negative influence on the model's output. `competitor_mentions` has the clearest pattern; high values of this feature push the model predictions

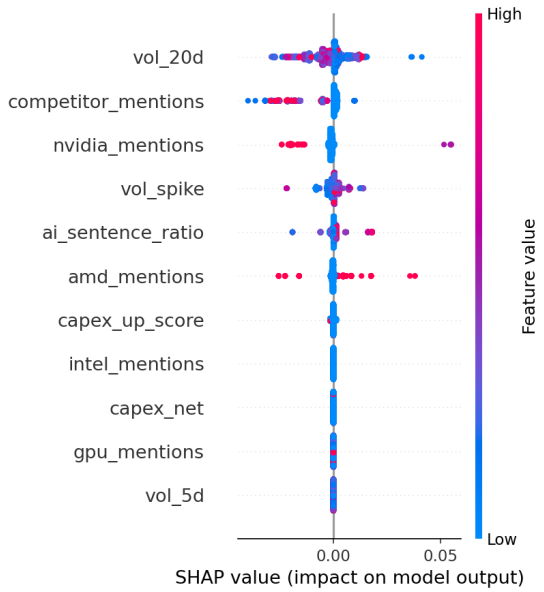


Figure 1: SHAP summary plot for the final-fold model. Red points indicate high feature values and blue points low values; points right of zero raise the predicted probability of outperformance.

down, indicating more discussion on competitors is often associated with lower predicted outperformance. `amd_mentions` and `nvidia_mentions` are also highly inconsistent and appear on both sides. Additionally, the `capex_down_score` feature scores a feature importance of zero, likely because the U.S. stock market is considered to be in a bull market, so capex reduction language is rare in recent SEC Filings. In other words, there is no single feature that is a significantly strong predictor of stock outperformance; rather, each feature contributes a small effect.

3.3 Sensitivity to Lookahead Leakage

As shown in Figure 2, the IC follows a positive trend when the label window is shifted to overlap the feature date by k trading days. $k = 0$ is the correctly constructed target (days $t+1$ to $t+5$), since there should be no lookahead bias; $k = 4$ signifies 4 past days were used as a factor of the target variable. On average, each extra day of leakage results in IC increasing by about 0.008. Since t=2.54, the slope is about 2.54 standard deviations above zero, so the upward trend is highly unlikely to be purely by chance. Since $p = 0.012 < 0.05$, the slope is significantly positive.

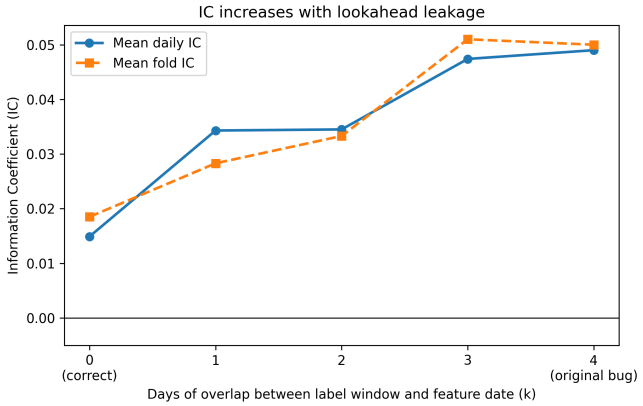


Figure 2: Out-of-sample IC rises as lookahead leakage increases.

4 Discussion

4.1 Limitations

A key limitation of this project lies in the data infrastructure when gathering news sentiment. First, the news articles retrieved through FNSPID could only be covered through 2023. This means we are unable to honestly test whether news mattered during the AI boom in 2024-2025, when we would expect AI coverage to be the most intense.

Additionally, when extracting SEC EDGAR 8-K Filings, three tickers (ASML, TSM, and ARM) each returned zero qualifying filings. This is because we explicitly filtered for issuers that file form 8-K, which is only required for US domestic companies, not accounting for 6-K material event disclosures, the equivalent of Form 8-K outside of the United States. However, ASML, TSM, and ARM, are foreign private issuers, from the Netherlands, Taiwan, and the United Kingdom, respectively.

Beyond coverage, the text features themselves are simple keyword counts based on dictionaries using case-insensitive substring matching. This approach easily ignores context and negation, and substring matching can produce false positives, such as 'h20' incorrectly matching to 'h200'. Model-based representations such as transformers may capture information these dictionaries miss.

Finally, this project only dealt with 19 tickers, significantly limiting the statistical power of cross-sectional ranking. The stock ticker universe was chosen based on companies' current associations with AI, meaning they are already likely to be AI-exposed and have more AI-terminology in their earnings calls and related news articles. This introduces lookahead bias in sample selection. Using more tickers and walk-forward folds (such as at least 20, rather than 5) would be expected to produce more reliable IC and ICIR estimates. Thus, the ICIR of 0.6671 should be interpreted with caution since

only 5 walk-forward folds were implemented.

4.2 Implications

Keyword-based features do not significantly improve cross-sectional prediction of five-day returns beyond volatility. This result holds even when news sentiment are added. A likely reason is that AI language no longer differentiates firms. After the release of ChatGPT, nearly every company in the sample began using AI terminology. This relates to the concept of 'AI-washing', a deceptive marketing tactic where companies promote their product or service by over-emphasizing the role and impact of artificial intelligence on it, misleading investors to be attracted to their business without necessarily having a substantial impact in the economy. In fact, a 2019 study found that 40% of new tech firms who claimed to be an 'AI start-up' were found to use "virtually no AI at all"(2). To further distinguish between these beneficiaries, in the future, we could implement context-aware NLP. By using tools such as "Named Entity Recognition" or dependency parsing, we could investigate whether the company is the one experiencing AI demand or if it is the object affected by other organizations' demand in AI. A second implication regards research practice. During feature engineering, an error arising from a single line of code initially inflated the model to have a seemingly significant signal of IC=0.061, which vanished once the error was corrected. The leakage analysis indicated that even a single day of overlap between the label and features is enough to turn a null result into a significant one. Since the baseline model was affected by the same leaked feature, it did not initially offer any warning of lookahead bias. Rather than blindly trusting validation design or baseline comparison, it is essential to also spot-check manually as well as write assertion checks that thoroughly cover edge cases.

4.3 Future Work

In future iterations of this project, we could strengthen the data infrastructure, text representation, and evaluation of the model.

Data Infrastructure: Since FNSPID news coverage stops in 2023 and only covers news headlines but not article content, we could extract data from GDELT or Tiingo News API. We would also extend the EDGAR coverage to include foreign private issuers by collecting Form 6-K Filings in order to ensure ASML, TSM, and ARM are also properly represented. Defining a larger and more relevant universe point-in-time, selecting stocks using information known as of each date rather than what are currently predominant in the AI industry in present-day, is also vital to removing lookahead bias in sample selection.

Text Representation: More fine-grained Regex text preprocessing could improve signal quality by filtering out boilerplate legal language, HTML artifacts, and financial tables that may interfere with text interpretation, in order to ensure `gpu_mentions` and `ai_sentence_ratio` accurately reflect genuine management commentary. Rather than counting AI terms in absolute terms, we could also measure how each firm's language changes relative to its own past filings, which Cohen, Malloy, and Nguyen show predicts future returns (4).

Subsequently, we could extend the model by comparing LightGBM against other transformer-based models, such as FinBERT, SEC-BERT (8), or Longformer (3).

Evaluation: We could extend the evaluation period to 2015-2025 to increase the number of test days and walk-forward folds, producing a more reliable ICIR estimate. Longer prediction horizons (e.g., 20–60 trading days) could test whether filing information is incorporated more slowly than five days allows.

5 Conclusion

This study tested whether AI-related language in SEC 8-K earnings releases adds predictive value beyond volatility when ranking AI-sector stocks over a five-day horizon. With our model, it does not. The model achieved a mean fold IC of 0.0185. The model's mean daily IC of 0.015 was not significantly different from zero (Newey-West $p=0.18$), and relative to a volatility-only model trained on the same folds, the text features added no statistically detectable value ($\Delta IC = 0.0082$, $p = 0.57$). This conclusion remained the same when news sentiment from FNSPID or standard momentum features were added. Features were dominated by `vol_20d`, the rolling volatility over a 20-day window. Early stopping retained a single tree, indicating little learnable structure in the text features. Aside from model limitations, one possible reason for no significant findings in the additions of AI-related language in SEC 8-K earnings is that AI terminology became nearly universal after 2022, so keyword counts no longer distinguish them between firms. Instead, we could measure how a company's language changes compared with its own past filings, rather than how much AI language it uses in absolute terms.

The study's second finding concerns methodology. An error in the original target construction produced an apparently strong signal ($IC = 0.061$) that disappeared once corrected. The leakage analysis showed that even one day of overlap between the label window and the feature date turns a non-significant result into a highly significant one ($p < 0.001$), with IC rising by roughly 0.008 per additional day of overlap ($p = 0.012$). Because the volatility baseline was inflated in parallel, outperforming it offered no warning. These results suggest that ap-

parent signals from financial text should be treated with caution until label construction has been verified directly, and that a carefully tested null result can be as informative as a positive one.

These findings should not be read as evidence that AI-related disclosure language contains no information. The null result applies to keyword-count features, a 19-stock universe, and a five-day horizon, and the design has limited statistical power: given the standard error of the daily IC, a true IC of roughly 0.03 would be needed for reliable detection, so a smaller but genuine effect could go undetected. Whether richer text representations and larger point-in-time samples can recover such an effect remains an open question.

Use of Generative AI

Claude (Anthropic) was used to assist with debugging analysis code, reviewing statistical methodology, and editing and formatting the manuscript. The author designed the study, ran all analyses, verified all results, and takes full responsibility for the content.

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